Slip 17

1)Define an Interface “Integer” with a abstract method check().Write a Java program to check whether a given number is Positive or Negative.

```
interface Integer {
  void check();
}
class NumberCheck implements Integer {
  int num;

  NumberCheck(int num) {
    this.num = num;
  }

  public void check() {
    if (num > 0)
      System.out.println(num + " is Positive");
    else if (num < 0)
      System.out.println(num + " is Negative");
    else
      System.out.println(num + " is Zero");
  }
}

public class PositiveNegative {
  public static void main(String[] args) {
    NumberCheck n = new NumberCheck(-10);
    n.check();
```

```
    }
}
class Cylinder implements Operation {
    double radius, height;
    Cylinder(double radius, double height) {
        this.radius = radius;
        this.height = height;
    }
    public void area() {
        System.out.println("Surface Area of Cylinder = "
                + (2 * PI * radius * (radius + height)));
    }
    public void volume() {
        System.out.println("Volume of Cylinder = "
                + (PI * radius * radius * height));
    }
}
public class OperationDemo {
    public static void main(String[] args) {
        Circle c = new Circle(5);
        c.area();
        c.volume();
       System.out.println();

        Cylinder cy = new Cylinder(3, 7);
        cy.area();
        cy.volume();
    }
```

```
}
```

2) Write a java program to demonstrate Synchronization Example.

```
class Table {
  synchronized void printTable(int n) {
    for (int i = 1; i <= 5; i++) {
      System.out.println(n * i);

      try {
        Thread.sleep(400);
      } catch (InterruptedException e) {
        System.out.println(e);
      }
    }
  }
}

class Thread1 extends Thread {
  Table t;

  Thread1(Table t) {
    this.t = t;
  }

  public void run() {
    t.printTable(5);
  }
}
```

```
class Thread2 extends Thread {
   Table t;

   Thread2(Table t) {
      this.t = t;
   }

   public void run() {
      t.printTable(10);
   }
}

public class SynchronizationDemo {
   public static void main(String[] args) {

      Table obj = new Table();

      Thread1 t1 = new Thread1(obj);
      Thread2 t2 = new Thread2(obj);

      t1.start();
      t2.start();
   }
}
```

OR

2) Write a program to create link list of integer objects. Do the following: